ORGANISATION – ENZYMES & DIGESTION

KEY VOCABULARY

Enzyme: a biological catalyst that speeds up reactions.

Substrate: the molecule an enzyme acts on.

Active site: the part of the enzyme the substrate fits into.

Denatured: the active site changes shape and stops working.

COMMON MISCONCEPTIONS

~~Enzymes are killed by high temperature.~~

✓ Enzymes are not alive — they are denatured.

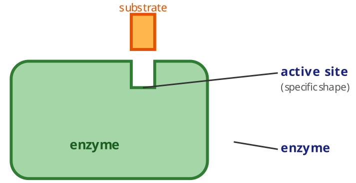
~~One enzyme works on any substrate.~~

✓ Enzymes are specific to one substrate shape.

~~Denaturing can be reversed by cooling.~~

✓ Denaturing is permanent — the shape is lost.

LOCK-AND-KEY MODEL



Only a substrate with the right shape fits the **active site**.

1 DEFINE THE TERM

Describe what is meant by an enzyme. **[2 marks]**

2 LOCK AND KEY

Explain why each enzyme only works on one type of substrate. **[2 marks]**

3 EFFECT OF TEMPERATURE

Explain what happens to an enzyme above its optimum temperature. **[3 marks]**

4 DIGESTIVE ENZYMES

Complete the table of enzymes using the word bank. **[2 marks]**

_____ breaks starch into _____ ; protease breaks protein into amino acids.

amino acids | glucose | protease | amylase

5 EFFECT OF PH

An enzyme works best at pH 7. Predict what happens to its rate at pH 3. **[2 marks]**

6 BILE

Explain how bile increases the rate of fat digestion. **[2 marks]**

TIP: *Substrate must fit the active site — like a key in a lock.*

✓ **Think: Why does the rate fall on BOTH sides of the optimum temperature?**